

Potential-Guided Planning in Magic Tower

A deterministic long-horizon resource-allocation benchmark

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Objective. Magic Tower is a fully observable deterministic RPG puzzle in which legality, combat damage, key economy, and event flags are all exactly simulable. We study whether route discovery can be improved by combining exact replay, potential-guided PUCT search, archive exploration, and a multi-agent reference planner.

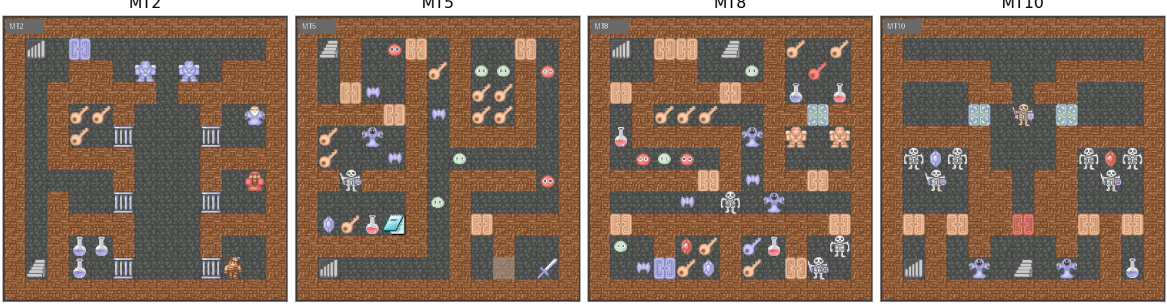
Strict replay outcomes

403
reference HP

315
PUCT HP

436
agentic HP

Game State as a Resource Graph



Floors expose a deterministic interaction graph: items, enemies, doors, stairs, merchants, and scripted events.

State and action abstraction

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}_{\text{valid}}, f, r, \gamma, \rho_0, \mathcal{G}, \mathcal{F}, H), \quad s_{t+1} = f(s_t, a_t).$$

Macro-actions select interaction nodes; walking and floor transfer paths are internal to the transition function.

Combat threshold

$$C(s, e) = \left\lceil \frac{\text{HP}_e}{\max(1, \text{ATK}(s) - \text{DEF}_e)} \right\rceil \max(0, \text{ATK}_e - \text{DEF}(s)).$$

A single ATK or DEF point can sharply reduce future damage, so reward and value estimates must represent marginal threshold effects.

Fast executable mask

The planner first computes the set of executable interaction nodes, then calls the simulator only for selected transitions.

- Static topology and interaction coordinates are precomputed once.
- Dynamic doors, defeated enemies, and consumed resources are encoded as compact blocker bitsets.
- Paths are recovered lazily only for the chosen macro-action.

MOTA-specific challenge

- Irreversible resource use:** keys, potions, and combat losses can make a much later floor infeasible.
- Threshold combat:** one ATK or DEF point can change multiple future enemy losses discontinuously.
- Hybrid state:** spatial topology, consumed map nodes, hero scalars, and event flags must be evaluated jointly.

Planner features

Hero resources, consumed-node bitsets, legal masks, missing keys, path length, current enemy damage, ATK+1/DEF+1 damage drops, boss margin, and unlocked resource value.

Core abstraction

The agent does not choose arrow keys. It chooses the next meaningful interaction node, while the simulator deterministically executes the shortest legal path.

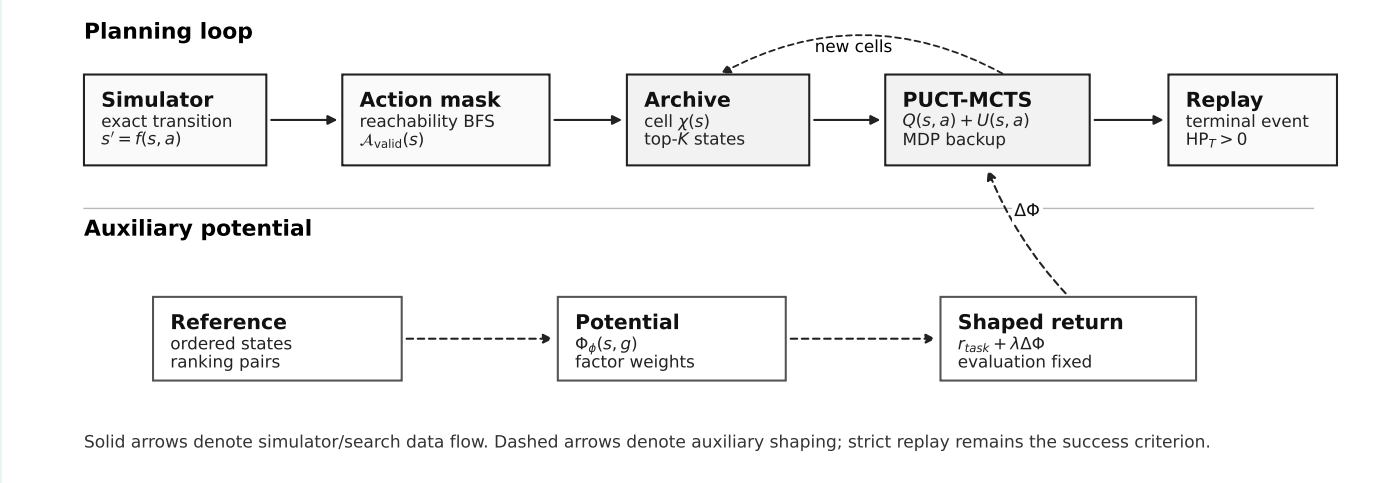
State signature

- Hero inventory:** HP, ATK, DEF, money, and three key counts.
- Map memory:** consumed resources, defeated enemies, opened doors, and event flags.
- Frontier information:** executable nodes, masked nodes, missing keys, damage margins, and newly unlocked resources.

Why graph actions help

The action space becomes the set of currently meaningful resource interactions, so search effort is spent on route choices rather than on repeated corridor movement.

Potential-Guided Planning Loop



The search loop uses exact simulator transitions, legal action masks, archive cells, PUCT rollouts, and strict replay validation.

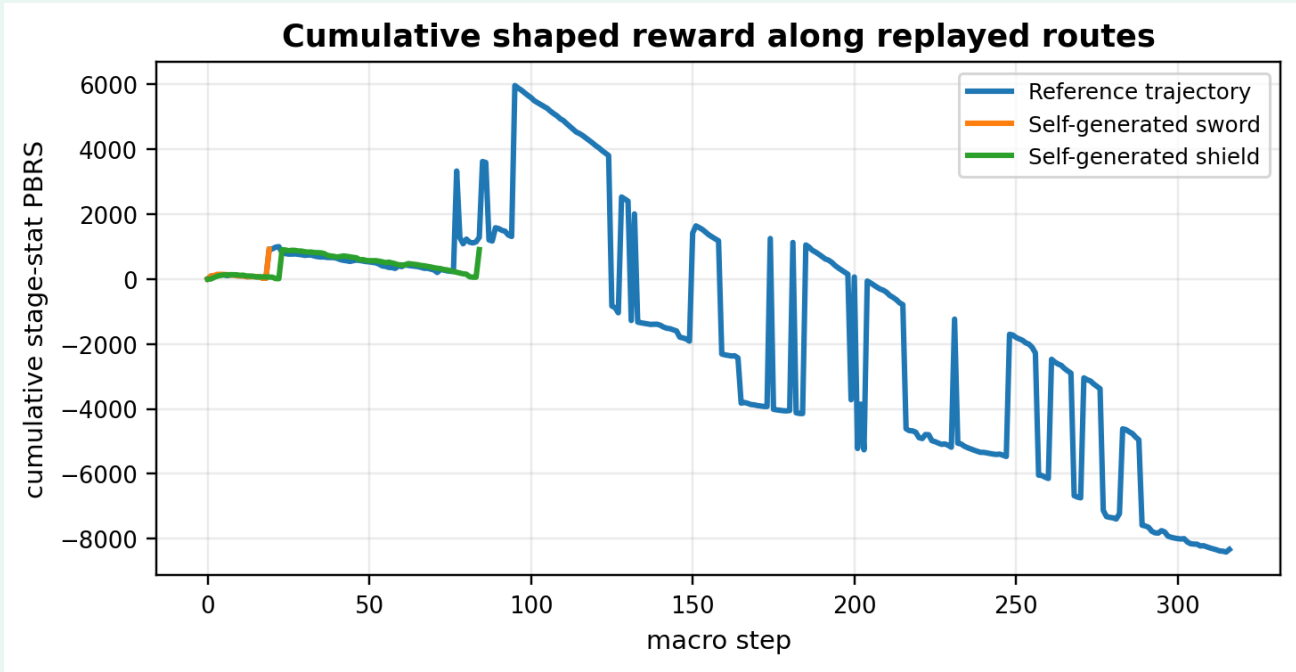
Complementary approaches

- Reference-route construction.** Ordered successful states validate the simulator and support progress estimation.
- AlphaGo Zero-style planning.** Policy/value priors guide single-agent PUCT-MCTS over macro-actions.
- Agentic planner.** Multiple scoring modules evaluate the same candidate after-states, while a beam arbiter keeps legal transitions.

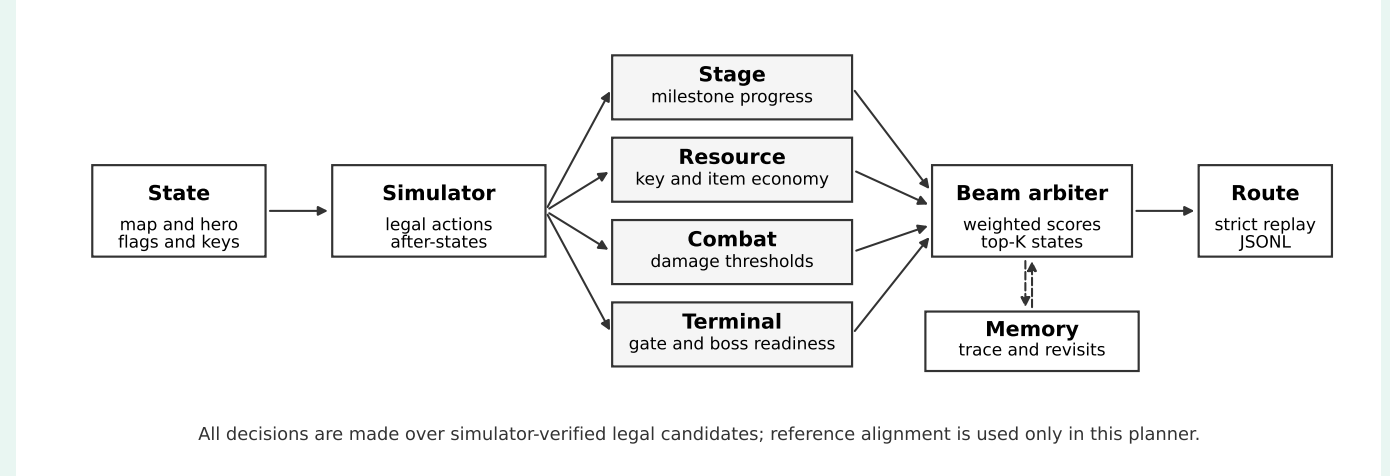
Potential-based shaping

$$r_{\text{search}} = r_{\text{task}} + \lambda [\gamma \Phi_{\phi}(s', g') - \Phi_{\phi}(s, g)].$$
$$\mathcal{L}_{\Phi} = - \sum_{(s^+, s^-)} \log \sigma(\Phi_{\phi}(s^+, g^+) - \Phi_{\phi}(s^-, g^-)).$$

The shaped reward guides search and ranking; strict terminal replay remains the evaluation objective.



Agentic reference planner



Archive-guided local search

$$\mathcal{B}[\chi(s)] = \text{top-}K \text{ non-dominated states in an abstract cell.}$$

Each iteration selects a promising archive cell, restores its simulator state, runs short PUCT rollouts, and inserts newly reached states by lexicographic score:

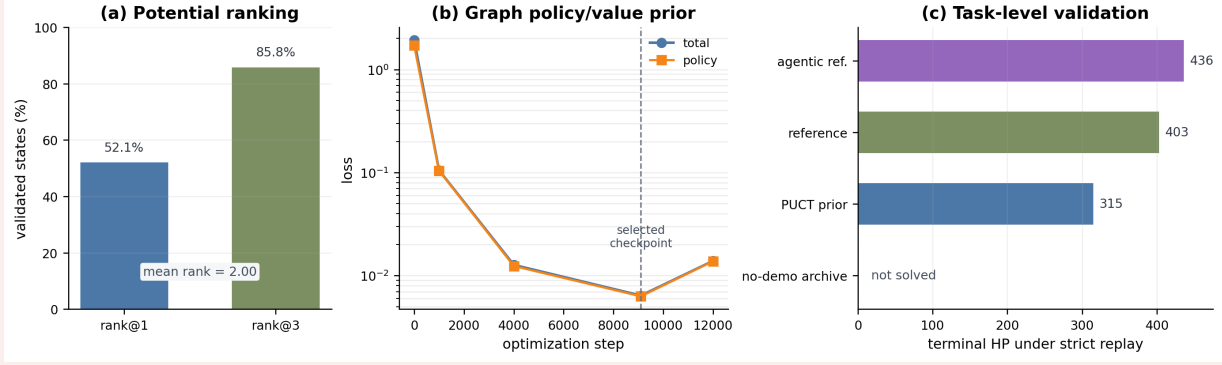
$$(1[\text{success}], \text{stage progress}, \text{HP}, \text{keys}, -\text{depth}).$$

$$\chi(s) = (g, \text{floor}, \text{ATK}_{bin}, \text{DEF}_{bin}, \text{keys}_{bin}, \text{HP}_{bin}, \text{boss}_{bin}).$$

Search principle

The archive preserves alternative resource schedules; MCTS then spends local computation near promising but not yet solved intermediate states.

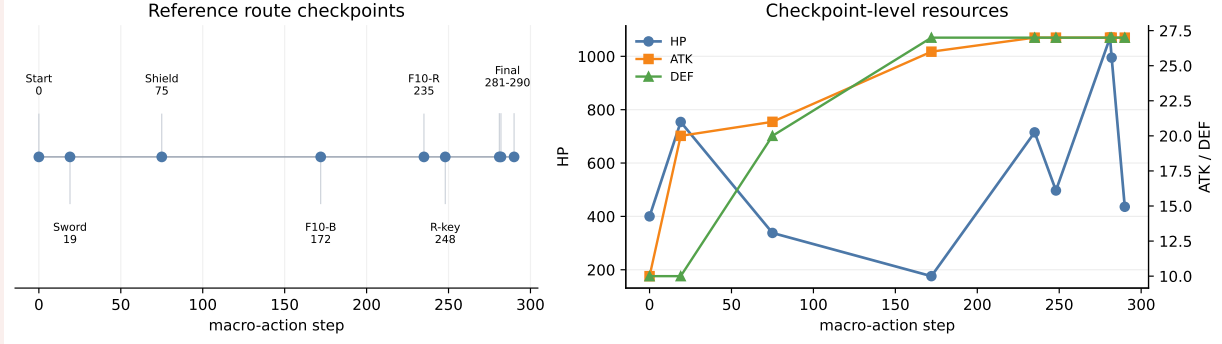
Evidence From Strict Replay



Potential ranking, policy/value fitting, and strict replay results are evaluated separately.

Strict replay confirms that all reported successful routes reach the terminal encounter with positive HP; the best multi-agent route improves the reference terminal margin.

Route structure



Successful play accepts controlled HP losses to acquire equipment and stat resources, then rebuilds terminal HP margin.

Single-agent PUCT backup

$$G_k = \sum_{i=k}^{L-1} \gamma^{i-k} r_{\text{search}}(s_i, a_i, s_{i+1}) + \gamma^{L-k} \hat{V}(s_L).$$

$$a = \arg \max_a Q(s, a) + c_{\text{puet}} P_{\theta}(a \mid s) \frac{\sqrt{\sum_b N(s, b)}}{1 + N(s, a)}.$$

Condition	HP	Interpretation
Reference route	403	simulator and potential-factor validation
PUCT prior	315	closed-loop search reaches terminal event
Agentic planner	436	stronger reference trajectory under strict replay
No-demo archive	—	reaches milestones but not terminal success

Evaluation discipline

- Strict replay is the validity criterion; shaped return is never counted as success.
- The potential model uses ordered successful states only as a dense search signal.
- Pure archive search is reported separately from the multi-agent reference planner; model confidence never replaces simulator replay.

Observed bottlenecks

Premature combat, key starvation, early potion use, local loops, and insufficient terminal HP explain most failed strict replays.

What the traces show

- Successful routes delay many combats until equipment and stat thresholds reduce deterministic losses.
- Key economy and terminal HP margin remain the binding constraints near the final encounter.

Benchmark Extension: Text, Video, and Symbolic Control

Agent-facing interface

A MOTA instance can be exposed as synchronized symbolic, textual, and visual observations while preserving exact transition validation.

Symbolic state, action masks, text prompts, and rendered route videos can be generated from the same deterministic transition log.

Why this benchmark scales

Community tower archives contain many hand-designed maps and play traces; these support both supervised pretraining and strict out-of-distribution evaluation.

Benchmark protocol

Input	symbolic state, text description, legal macro-actions, or rendered video.
Action	choose one legal interaction node; the simulator executes the deterministic path.
Validity	exact replay under the transition log, not model confidence or shaped return.
Quality	terminal success, terminal HP, route length, and transfer to modified towers.

This protocol connects symbolic planning, language-conditioned control, and video-based agent evaluation without changing the underlying transition function.

Evaluation axes

Validity is exact replay; quality is terminal HP and route length; coverage is the number of distinct milestone schedules; learning is transfer across modified maps and unseen towers.

Open bottlenecks

- sparse terminal reward under-covers late key and HP schedules;
- potential learning improves progress ranking, but cannot replace archive diversity;
- text/video agents must explain both legality and long-horizon resource trade-offs.

Takeaways and released artifacts

Exactness first

All reported routes are checked by deterministic replay, so terminal HP, key balances, defeated enemies, and event flags are auditable rather than inferred from model score.

Learning as guidance

The potential model supplies dense progress information for PUCT and archive priority; the terminal objective remains strict replay success with positive HP.

Benchmark trajectory

The same environment supports symbolic agents, language-conditioned agents, and video-based agents over a shared replayable action interface.

Transition log schema: checksum $\rightarrow$ action mask $\rightarrow$ interaction node $\rightarrow$ combat/key/event update $\rightarrow$ potential factors $\rightarrow$ strict replay verdict.

Simulator
deterministic rules

Graph state
all-node action mask

PUCT-MCTS
policy-value search

Potential
progress shaping

Visualizer
text-video replay

Artifacts: simulator, macro-action environment, graph-state encoder, potential training logs, archive cells, trajectory logs, failure cases, route visualizer, and report figures.

Central claim: exact symbolic replay gives a hard success criterion, while potential-guided PUCT and archive search provide scalable guidance for long-horizon resource planning.